

Seven questions about the cell

I'm building a simulation that mirrors the real cell on screen while it runs. It **only reads** — it never sends a command to the PLC or to either robot. These seven answers tell me where to read from and what the readings mean. Short answers are fine; "I don't know" is a useful answer too.

These three change how I build it, so they're the ones I'd like first.

1

Is there a network switch at the bench that the PLC and both robots plug into? Or is each thing wired on its own?

My laptop has to sit on the same network as them to read anything at all.

2

When you program the PLC, do you plug in an Ethernet cable or a USB cable?

If it's always USB, the PLC's network side may never have been switched on. That's not a problem — it just becomes a small setup job, and it's better to know now than on the day.

3

What turns the vacuum on — the PLC, or the Pro 600 robot's own outputs?

Changes how I build it. My code assumes the PLC controls it. If the robot does, I've been planning to read the wrong device.

4

Do the PLC and the two robots talk to each other over the network, or are they joined by plain wires — one sends a "go" signal and the other reacts?

Changes how I build it. It decides whether the PLC knows what the robots are doing, or whether I have to read all three separately and line them up myself.

5

Who starts a cycle — somebody pressing a button, the PLC on its own, or a person hitting play on a robot pendant?

Tells me what signal to watch for as "a new cycle has begun", so the simulation starts at the same moment the machine does.

6

Are there any sensors on the cell at all — something that detects the holder arriving, or that a wafer is present?

Changes how I build it. Right now I assume there are none and work out where the holder is by counting seconds from when the belt starts. A real sensor would be far more accurate, and I'd rather read it than guess.

7

Does the belt ever run backwards, or does the holder always travel one way and get carried back by hand?

In my simulation the belt returns the holder by itself. If the real one doesn't, the two stop matching after the first cycle.